PRODUCTION ARCHITECTURE • VERSION 1.0

NIVASA

Technical Design Document — Residential Society Management Platform

GitHub Repository: https://github.com/sumanpatra006/nivasa.git

Live Web App: https://nivasa.vercel.app

Target Scale: 2,400 Apartments • 28 Towers

Date: November 2026

Executive Summary

Nivasa is an enterprise residential operating platform designed for Indian cooperative housing societies and high-rise apartment complexes. It replaces fragmented paper registers, disparate spreadsheets, and unmonitored WhatsApp groups with a unified, high-integrity system.

The platform addresses five foundational societal workflows: automated maintenance billing with an absolute integer-paise standard (zero floating-point drift), deterministic 7-evidence bank statement reconciliation, hybrid AI grievance triage with deterministic life-safety emergency overrides, DPDP Act 2023–compliant gatekeeper kiosk kiosks with offline synchronization, and an executive committee oversight dashboard.

1. System Architecture & Technology Justification

1.1 The Modular Monolith

Nivasa is architected as a **Modular Monolith** structured under a single monorepo (`@nivasa/shared` , `@nivasa/backend` , `@nivasa/frontend`).

> **Why Modular Monolith Over Microservices:**
> Residential society finance requires atomic relational consistency across invoices, receipts, and flat ledgers. A distributed microservices architecture introduces distributed transaction failure modes (two-phase commit latency, saga rollbacks), which are unacceptable for audited housing society records. A modular monolith guarantees ACID guarantees inside PostgreSQL transactions while preserving clear domain boundaries.

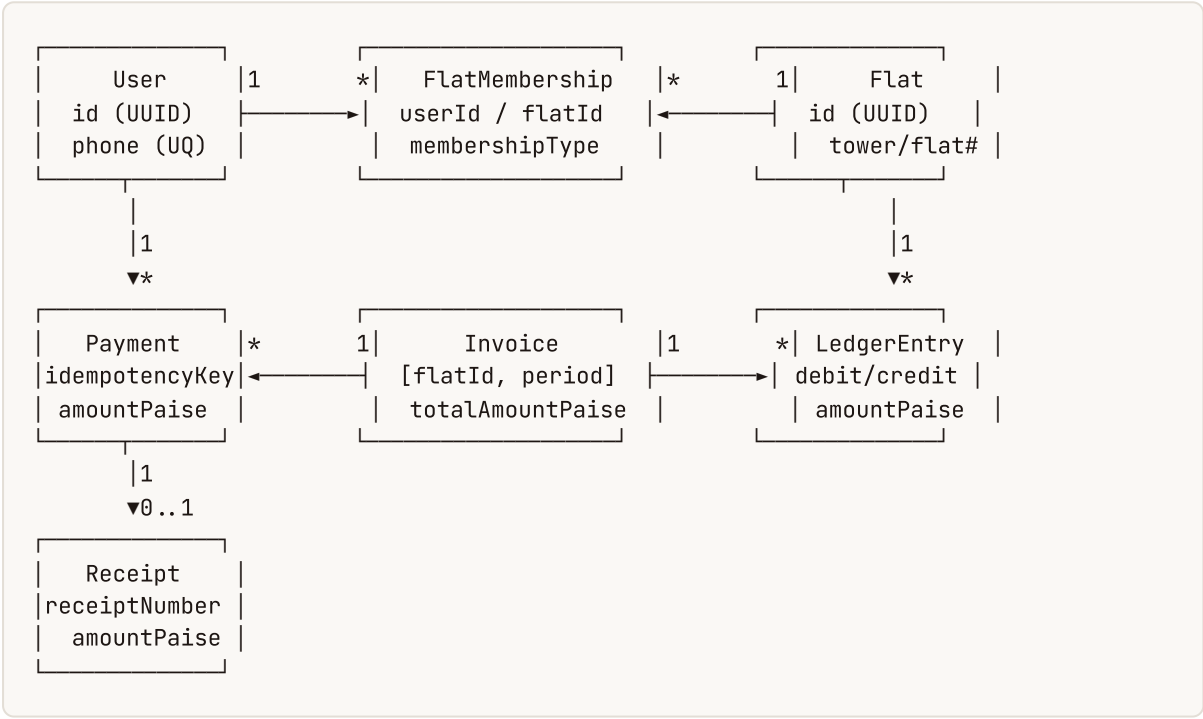
Layer	Technology	Architectural Rationale
Language	TypeScript 5.x (Strict)	End-to-end type safety spanning database entities, REST responses, and UI forms.
Frontend	React 18, Vite, Tailwind CSS	Sub-second HMR, high rendering performance for large bank reconciliation queues, and responsive tablet UI.
State / Data	TanStack Query & Zod	Automatic background cache invalidation and shared schema validation across frontend and backend.
Backend API	Node.js, Express	Asynchronous non-blocking I/O with modular routers, middleware RBAC, and streaming CSV parser.
Database	PostgreSQL (Neon) + Prisma ORM	Strong relational integrity, foreign key cascading, unique transaction indices, and connection pooling.

2. Relational Data Model & Financial Mathematics

2.1 The Integer-Paise Standard

Financial calculations never use JavaScript `Number` floating-point arithmetic. All balances, late fees, and rates are stored as **integer paise** (1 INR = 100 paise). An 1,150 sq ft flat billed at ₹6.50/sq ft computes as `1150 × 650 = 747,500` paise (₹7,475.00), preventing any fractional rounding loss.

2.2 Core Relational Schema



3. Application Workflows & AI Implementation

3.1 Bank Statement Reconciliation Engine

Bank statements (HDFC, ICICI, SBI CSV exports) are parsed with zero silent row discards. A deterministic 7-evidence matching algorithm evaluates candidate invoices:

- **Tier 1 (+40%):** Exact payment reference string match in bank narrative.
- **Tier 2 (+25%):** Verified resident UPI ID match.
- **Tier 3 (+20%):** Tower and flat identifier match (e.g. "T07-1203").
- **Tier 4 (+15%):** Exact amount matching invoice outstanding balance.
- **Tier 5 & 6 (+10% each):** Due-date proximity and past resident payment method.

- **Ambiguity Protection:** If multiple candidate flats match the same score or amount, automatic posting is halted, and the transaction is placed into `NEEDS_REVIEW` for human accounting review.

3.2 Complaint AI Triage & Deterministic Safety Escalation

Grievances undergo a dual-stage safety and intelligence pipeline:

> **Deterministic Safety Rule Engine (Priority 1):**
> To prevent dangerous AI hallucinations, complaints mentioning *"gas smell"*, *"sparks / fire"*, *"lift stuck with passenger"*, or *"electric shock"* immediately override AI classification, assign **CRITICAL** urgency, dispatch an alert to the facility duty manager, and lock out automated closure.

For non-emergency complaints, the **Google Gemini AI Adapter** categorizes the issue (Plumbing, Electrical, Lift, Security, etc.), infers urgency, and recommends a maintenance technician with transparent reasoning. If external AI services are offline, a local keyword fallback engine seamlessly assumes triage duties with no disruption.

3.3 Gatekeeper Tablet & DPDP Act 2023 Compliance

- **Touchscreen Ergonomics:** High-contrast 64px touch targets and dedicated PIN keypad designed for low-cost outdoor gate tablets.
- **Offline Queue:** Local IndexedDB queue buffers arrivals when society Wi-Fi drops and synchronizes idempotently via client UUIDs upon reconnection.
- **90-Day Privacy Purge:** Physical visitor photo files are permanently expunged from disk after 90 days in compliance with statutory Indian data minimization laws.

4. UI/UX Design Decisions

Nivasa departs from generic, harsh SaaS aesthetics in favor of a **Warm Editorial Design System**:

- **Color Palette:** Deep Walnut (`#2c221e`) and Warm Sand (`#faf8f5`) establish an inviting, authoritative atmosphere that minimizes glare for elderly residents.
- **Visual Density:** High-density data tables with monospaced numbers (`tabular-nums`) for accounting staff; spacious, friendly card layouts for residents.
- **Accessibility:** WCAG AA 4.5:1 contrast compliance across all text elements and interactive form controls.

5. Scope Adjustments & Future Roadmap

5.1 What Was Deliberately Cut

- **Video Intercom via WebRTC:** Cut in favor of fast 6-digit PIN passes to prevent gate queue congestion.
- **Automated License Plate Recognition (ANPR):** Cut due to high hardware camera dependency and non-standard Indian plate fonts.
- **Full Double-Entry Balance Sheet General Ledger:** Flat ledgers and dues ageing were prioritized over depreciation schedules.

5.2 Future Roadmap (Phase 2)

- **Live Dynamic UPI QR Webhooks:** Real-time payment confirmation via Razorpay/Cashfree webhook callbacks.
- **Multi-Gate WebSocket Cluster:** Instant sync across North, South, and Service gates using Redis pub/sub.
- **Multilingual Voice Dictation:** Speech-to-complaint logging in Hindi and Marathi using Whisper and Gemini.
- **IoT Sub-Meter Telemetry:** Automated digital water and DG power meter logging into monthly invoices.